

G1 Physical Campaign — Interim Report (Session 1)

2026-07-08, real laboratory · 8 runs (6 baseline + 2 governed) · session paused ~11:10, to be resumed

1. Executive summary

The first hardware session produced the two most important single data points of the entire evaluation. (i) The **ungoverned baseline reproduced the historical failure mode**: run 6 wedged into a door-frame collision loop for 513 s (7 collisions, 50 m of thrashing) until manually killed. (ii) The **governed system demonstrated the full HELP chain on hardware**: with a brimful cup, real human spill marks drove the fragility meta-parameter into its dangerous region, the hard veto emptied the deployable set, sustained HELP triggered the experience-escalation layer, and the robot stopped itself in 23 s — a governed, bounded, self-diagnosed stop versus an 8.5-minute human-supervised failure. This is the paper's conservative non-selection semantics with physical ground truth. The practical consequence — a brimful cup (1 cm freeboard) cannot survive the start-manoeuvre turns under any speed governance — is itself a finding (turn rate is not yet part of the analogy profile) and sets the protocol for session 2 (200 g fill for completed governed runs).

2. Runs completed

Run	Arm / leg	Result	Time (s)	Spill marks	Collisions	Water
102149	R1 baseline A→B	reached	74.6	14	0	246 g fill
102421	R1 baseline B→A	reached	48.4	3	0	→ 116 g (RT: 130 g)
103218	R1 baseline A→B	reached	99.7	18	2 (IMU grazes)	246 g fill
103558	R1 baseline B→A	reached	124.3	8	0	→ 96 g (RT: 150 g)
104244	R1 baseline A→B	reached	106.0	27	0	246 g fill
104520	R1 baseline B→A	MANUAL ABORT	513	3	7 (door frame)	→ 77 g (RT: 169 g)
110018	R2 governed A→B	HELP abort (governed)	23.0	4 (t=10-14 s)	0	246→162 g
110554	R2 governed A→B	HELP abort (governed)	14.6	4 (t=3.5-6.7 s)	0	246→209 g

Baseline aggregate (6 runs): 5/6 reached, **12.2 spill marks/run**, ~449 g of water lost in total (5.6-7.6 g per marked event), 9 collisions, reached-run times 48-124 s (mean ≈ 91 s). Fresh-cup front-loading (most spills in the first ~25 s) reproduced in 3 of 3 fresh fills.

3. Real robot vs digital twin

Quantity	Digital twin (82 runs)	Real robot (session 1)	Reading
Baseline time A→B	83.4 ± 5.2 s	48-124 s, mean ≈ 91 s	Consistent; real variance larger
Baseline spills/run	0.50 ± 0.37	12.2 marks/run (~75 g/run)	Real gait ≈ 20× more excitatory than modelled: K_GAIT recalibration datum. Direction of the governed contrast unchanged
Water per spill event	FB_LOSS = 4 mm column (assumed)	5.6-7.6 g ≈ 1.4-1.9 mm column	Model constant to be re-fit (≈ 1.5 mm)
Fresh-cup front-loading	Predicted (freeboard mechanism)	Reproduced 3/3 fresh fills	Mechanism validated on hardware
Baseline failure mode	No door loop in twin (no clutter, wider twin door)	513 s door-frame loop, 7 collisions, manual kill	The historical failure (run 100927-style) is real and ungoverned; motivates escalation
HELP semantics	3/8 single-candidate arm aborts (mobility veto)	2/2 governed fresh-cup runs: spill marks → fragility veto → HELP → stop in 23.0 / 14.6 s	Full chain confirmed on hardware with human ground truth — first such demonstration
Start-phase spills	Present but below spill threshold (smooth planar start)	3-4 marks in first 15 s of EVERY fresh-cup run, governed or not	NEW finding: start turns dominate; turn rate is not governed (only forward speed) — post-campaign work item: angular-speed in analogy profiles
Governed completed runs	8/8 reached, 0 spills	None yet (2 HELP aborts at 246 g fill)	Session 2: 200 g fill ("payload level B") + 2 baseline runs at 200 g for comparability

4. Theory validation status after session 1

Confirmed on hardware: (a) conservative non-selection / HELP semantics with real human ground truth feeding the fragility channel — the single most important governance claim; (b) the spill model's freeboard mechanism (front-loading) and jerk-dominance (start turns, not cruise speed alone); (c) the ungoverned failure mode that experience escalation exists to prevent, timed at 513 s versus 23 s governed. **Pending (session 2):** the headline availability-vs-deployability trade-off needs completed governed runs (200 g protocol); the Layer-2 cross-run recovery (wrong-prior arm R3) and the sealed-lid deltas (R4) are untouched. **New knowledge produced by hardware:** turn manoeuvres are the dominant start-phase spill source and are currently outside the governed profile — an extension (angular-rate ceilings per analogy, mirroring the thesis profiles' multi-parameter policies) is queued for after the campaign.

5. Session 2 checklist

(1) Robot relocalised at a waypoint; perception + proxy + marker terminals up. (2) R2 remaining runs at **200 g fill**, same command and state file (meta2_state_real_m2.json — REMEMBER to include it in git commits; it already holds two runs of mismatch evidence). (3) Then R3 wrong-prior (fresh meta2_state_real_wrong.json, 6 runs, note PI after each). (4) R4 sealed lid (4 runs). (5) Two baseline runs at 200 g for the fill-level pairing. Marking discipline: a continuous slosh burst = ONE mark; ≥ 2 s calm separates events.

Data: all 8 runs pushed to main (datasets + filmstrips + goto.log). Companion documents: Digital-Twin Experiment rev. 4, Real-Robot Campaign Guide.